

ArithmeticSite

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Chapter 1

The multiplicative monoid $\mathbb{N}^\times$

The Arithmetic Site, introduced by Connes and Consani, is a geometric object — a topos equipped with a structure sheaf — whose points in the topos-theoretic sense are in natural bijection with the adèle class space of $\mathbb{Q}$. Its construction begins with a single, elementary monoid: the positive integers under multiplication. Everything else — the tropical semiring, the presheaf topos, the structure sheaf, and ultimately the connection to the Riemann zeta function — is built on top of this one piece of data.

Definition 1.1. Let $\mathbb{N}^\times$ denote the set $\{1, 2, 3, \dots\}$ of strictly positive integers equipped with ordinary multiplication. This is a commutative monoid with unit 1. In Lean we represent this as `PNat` with its multiplicative structure, via the abbreviation `NPos`.

The unique factorisation theorem says that $\mathbb{N}^\times$ is the free commutative monoid on the set of primes: every element factors uniquely as a finite product $\prod_p p^{a_p}$. This arithmetic richness — the interplay of primes, divisibility, and factorisation — is what makes $\mathbb{N}^\times$ the right monoid for the Arithmetic Site. As we will see, it is ultimately responsible for the fact that the points of $\widehat{\mathbb{N}^\times}$ are parametrised by the adèle class space of $\mathbb{Q}$, and that the Hasse–Weil zeta function of the site recovers the Riemann zeta function.

Definition 1.2. The *classifying category* $B\mathbb{N}^\times$ of $\mathbb{N}^\times$ is the one-object category whose single object is $*$ and whose morphisms $*$ $\rightarrow$ $*$ are the elements of $\mathbb{N}^\times$, with composition given by multiplication.

Remark 1.3. The passage from a monoid M to the one-object category BM is the standard *delooping* construction. A presheaf on BM — that is, a functor $BM^{\text{op}} \rightarrow \mathbf{Set}$ — is exactly a set X equipped with a left action of M : each element $n \in M$ acts as a function $X \rightarrow X$, and the action is compatible with composition. (Left and right actions coincide here since $\mathbb{N}^\times$ is commutative.) The presheaf topos $\widehat{B\mathbb{N}^\times}$ introduced in Chapter 3 is precisely the category of such sets-with- $\mathbb{N}^\times$ -action, and it is in this topos that the Arithmetic Site lives.

Chapter 2

The tropical semiring $\bar{\mathbb{N}}$

The tropical semiring $\bar{\mathbb{N}}$ is the second fundamental ingredient of the Arithmetic Site. It plays the role of the *structure sheaf*: as Connes and Consani put it, $\bar{\mathbb{N}}$ is the semiring on which $\mathbb{N}^\times$ acts by multiplication, and it is this action that makes $\bar{\mathbb{N}}$ into a sheaf of semirings over the topos $\widehat{B\mathbb{N}^\times}$. The arithmetic of $\bar{\mathbb{N}}$ is characteristic-1 arithmetic — addition is idempotent — which is the tropical, or min-plus, shadow of ordinary arithmetic. Connes and Consani define $\bar{\mathbb{N}}$ explicitly as $(\mathbb{N} \cup \{+\infty\}, \inf, +)$, confirming the min-plus convention: addition is min, with $+\infty$ as the additive identity. When elements of $\bar{\mathbb{N}}$ are written as exponents q^x for $q \in (0, 1)$, the addition becomes max on values (since smaller exponent means larger value), which is why the stalks in the paper appear in max-plus form as $[0, 1]_{\max}$ and $\mathbb{R}_{\max}$.

Definition 2.1. The *tropical semiring* $\bar{\mathbb{N}}$ is the set $\mathbb{N} \cup \{+\infty\}$ equipped with:

- *Tropical addition*: $a \oplus b := \min(a, b)$, with identity element $+\infty$.
- *Tropical multiplication*: $a \odot b := a + b$ (ordinary addition), with identity element 0.

In Lean, $\bar{\mathbb{N}}$ is represented as `Tropical (WithTop ℕ)`, via the abbreviation `NBar`. The zero element is `trop ⊤` (i.e. $+\infty$) and the multiplicative identity is `trop 0`.

Lemma 2.2. $\bar{\mathbb{N}}$ is a commutative semiring.

Proof. This is the Mathlib instance `CommSemiring (Tropical (WithTop ℕ))`, found by `inferInstance`. □

Lemma 2.3. Tropical addition on $\bar{\mathbb{N}}$ is idempotent: $a \oplus a = a$ for all $a \in \bar{\mathbb{N}}$.

Proof. $a \oplus a = \min(a, a) = a$. In Lean this follows from `Tropical.add_eq_left`. □

Remark 2.4. Idempotency of addition — equivalently, the identity $1 \oplus 1 = 1$ — is the defining feature of *characteristic 1* arithmetic. It is what places $\bar{\mathbb{N}}$ in the same conceptual family as finite fields $\mathbb{F}_p$ (characteristic p), with min playing the role that $+\text{ mod } p$ plays there. The scaling endomorphisms defined below are the characteristic-1 analogue of the Frobenius endomorphism $x \mapsto x^p$ of $\mathbb{F}_p$.

Definition 2.5. For each $n \in \mathbb{N}^\times$, the *scaling endomorphism* $\varphi_n : \bar{\mathbb{N}} \rightarrow \bar{\mathbb{N}}$ is defined by

$$\varphi_n(x) := nx \quad (x \in \mathbb{N}), \quad \varphi_n(+\infty) := +\infty,$$

where nx denotes ordinary multiplication in $\mathbb{N}$. Equivalently, $\varphi_n(x) = x^{\odot n}$, the n -fold tropical product of x with itself. In Lean, φ_n is defined as a semiring endomorphism $\mathbb{N}\text{Bar} \rightarrow^{+*} \mathbb{N}\text{Bar}$ via `scalingEndomorphism`.

Lemma 2.6. *For each $n \in \mathbb{N}^\times$, the map φ_n is a semiring endomorphism of $\bar{\mathbb{N}}$: it preserves both operations and both identities.*

Proof. Preservation of $+\infty$ and of tropical multiplication are immediate. Preservation of tropical addition uses $n \cdot \min(a, b) = \min(na, nb)$ for $n > 0$, which holds because scaling by n is order-preserving on $\mathbb{N} \cup \{+\infty\}$. In Lean this is the `map_add'` field of `scalingEndomorphism`, proved via monotonicity of `nsmul` and a case split on `le_total`. $\square$

Lemma 2.7. *The scaling endomorphisms compose multiplicatively:*

$$\varphi_m \circ \varphi_n = \varphi_{mn}$$

for all $m, n \in \mathbb{N}^\times$. In particular, the map $n \mapsto \varphi_n$ is a monoid homomorphism from $(\mathbb{N}^\times, \times)$ into the semiring endomorphism monoid $\text{End}_{\text{Semiring}}(\bar{\mathbb{N}})$.

Proof. $(\varphi_m \circ \varphi_n)(x) = m(nx) = (mn)x = \varphi_{mn}(x)$. In Lean this is `scalingEndomorphism_comp`, proved via `pow_mul`. $\square$

Remark 2.8. The Frobenius correspondences $\Psi(\lambda)$, parametrised by $\lambda \in \mathbb{Q}_+^\times$, are a more refined object: they live in the square $\widehat{B\mathbb{N}^{\times 2}}$ of the Arithmetic Site and are defined via the action of $\mathbb{N}^\times \times \mathbb{N}^\times$ on $\bar{\mathbb{N}} \otimes_{\mathbb{B}} \bar{\mathbb{N}}$. They are introduced in Chapter 7.

Chapter 3

The presheaf topos $\widehat{\mathbb{N}^\times}$

The topos underlying the Arithmetic Site is the simplest kind: a presheaf topos. Its objects are sets equipped with a left $\mathbb{N}^\times$ -action by arbitrary functions — not necessarily bijections, since $\mathbb{N}^\times$ is a monoid rather than a group — and its morphisms are $\mathbb{N}^\times$ -equivariant maps between them. The multiplicative structure of $\mathbb{N}^\times$ is entirely encoded in the composition of these actions: if n and m act on a set X , then nm acts as the composite $x \mapsto n \cdot (m \cdot x)$.

Connes and Consani note that $\widehat{\mathbb{N}^\times} \simeq \text{Sh}(\mathbb{N}^\times, J)$, where J is the *chaotic topology* on $\mathbb{N}^\times$ — the topology in which every presheaf is already a sheaf. This means there are no genuine gluing conditions to check: the topos is simply the presheaf category, with no further data. The covering families arising from factorizations of integers, foreshadowed in Chapter 1, reflect a finer site structure that one *can* put on $\mathbb{N}^\times$; the Arithmetic Site itself uses the coarser chaotic topology, where that combinatorial structure is not needed.

Definition 3.1. The *presheaf topos* $\widehat{\mathbb{N}^\times}$ is the category of functors $B\mathbb{N}^{\times\text{op}} \rightarrow \mathbf{Set}$, i.e. the category of sets equipped with a left $\mathbb{N}^\times$ -action. In Lean this is `Action (Type*) (Multiplicative ℕ+)`.

Lemma 3.2. $\widehat{\mathbb{N}^\times}$ is a Grothendieck topos.

Proof. Any presheaf category on a small category is a Grothendieck topos. The site is $(\mathbb{N}^\times, J)$ where J is the chaotic topology. $\square$

Remark 3.3. The points of $\widehat{\mathbb{N}^\times}$ — geometric morphisms from $\mathbf{Set}$ to $\widehat{\mathbb{N}^\times}$ — are the deep link to number theory. Connes and Consani prove (Theorem 2.2 of their 2014 paper) that the category of points of $\widehat{\mathbb{N}^\times}$ is canonically equivalent to the category of totally ordered groups isomorphic to non-trivial subgroups of $(\mathbb{Q}, \mathbb{Q}_+)$. The space of isomorphism classes of such points is canonically isomorphic to the double quotient

$$\mathbb{Q}_+^\times \backslash \mathbb{A}^f / \widehat{\mathbb{Z}}^*,$$

where $\mathbb{A}^f$ is the ring of finite adèles of $\mathbb{Q}$ and $\widehat{\mathbb{Z}}^* = \widehat{\mathbb{Z}}^\times$ is the group of units of the profinite completion of $\mathbb{Z}$. This quotient is a piece of the adèle class space of $\mathbb{Q}$, the same space that appears in Connes’ noncommutative-geometric approach to the Riemann hypothesis. The structure sheaf $\mathcal{O} = \widehat{\mathbb{N}}$ then equips this space with characteristic-1 geometry, whose Hasse–Weil zeta function recovers the Riemann zeta function.

Chapter 4

The structure sheaf

In classical algebraic geometry, a *ringed space* is a topological space equipped with a sheaf of commutative rings — the *structure sheaf* — whose stalks encode the local algebra of the space. The Arithmetic Site is a *ringed topos* in the same spirit: the role of the topological space is played by the presheaf topos $\widehat{\mathbb{N}^\times}$, and the role of the structure sheaf is played by $\bar{\mathbb{N}}$, viewed as a semiring object living inside that topos.

An object of $\widehat{\mathbb{N}^\times}$ is a set equipped with a left $\mathbb{N}^\times$ -action, so for $\bar{\mathbb{N}}$ to be a semiring *object* of $\widehat{\mathbb{N}^\times}$ the action must be by semiring endomorphisms: each scaling map φ_n must preserve both tropical operations and both identities. This is exactly the content of Lemma 2.6, which is therefore the key structural fact underlying the definition below.

Definition 4.1. The *structure sheaf* $\mathcal{O}$ of the Arithmetic Site is the object of $\widehat{\mathbb{N}^\times}$ whose underlying set is $\bar{\mathbb{N}}$ and whose $\mathbb{N}^\times$ -action is given by the scaling endomorphisms: $n \cdot x := \varphi_n(x) = nx$.

Lemma 4.2. The assignment $n \mapsto \varphi_n$ defines a monoid homomorphism $(\mathbb{N}^\times, \times) \rightarrow \text{End}_{\text{Semiring}}(\bar{\mathbb{N}})$, making $\bar{\mathbb{N}}$ into a semiring object of $\widehat{\mathbb{N}^\times}$ with $\mathbb{N}^\times$ acting by multiplication.

Proof. Each φ_n is a semiring endomorphism by Lemma 2.6. The homomorphism property $\varphi_{mn} = \varphi_m \circ \varphi_n$ is Lemma 2.7, and $\varphi_1 = \text{id}$ since $\varphi_1(x) = 1 \cdot x = x$. In Lean this is `scalingEndomorphism_comp`. $\square$

Remark 4.3. Since the Arithmetic Site uses the chaotic topology, every presheaf on $B\mathbb{N}^\times$ is automatically a sheaf — there are no gluing conditions to verify. The content of the structure sheaf is therefore entirely algebraic: $\bar{\mathbb{N}}$ is not merely a set with $\mathbb{N}^\times$ -action, but a semiring whose operations are *preserved* by that action. This is what Lemma 4.2 records.

Remark 4.4. Connes and Consani (Theorem 2.4 of their 2014 paper) describe the stalks of $\mathcal{O}$ explicitly. At the point corresponding to an intermediate semifield $F \subset K \subset \bar{F}$, the stalk is

$$\mathcal{O}_K = \{r \in K \mid r \oplus 1 = 1\},$$

the subsemiring of elements that are “bounded above by 1” in the ordering of K . This is the characteristic-1 analogue of the ring of integers of a local field: just as the ring of integers of a p -adic field K_v consists of elements of norm at most 1, the stalk of $\mathcal{O}$ picks out the elements “bounded above by 1”. At the “generic” point (corresponding to $K = F = Z_{\max}$) the stalk recovers $\bar{\mathbb{N}}$ itself.

Chapter 5

The Arithmetic Site

All the ingredients are now in place. The presheaf topos $\widehat{\mathbb{N}^\times}$ provides the geometric stage: a Grothendieck topos whose points are parametrised by the adèle class space of $\mathbb{Q}$. The tropical semiring $\bar{\mathbb{N}}$, equipped with its $\mathbb{N}^\times$ -action by scaling endomorphisms, is a semiring object of $\widehat{\mathbb{N}^\times}$: it plays the role of the ring of functions, recording characteristic-1 algebra at every point. Their combination is the Arithmetic Site.

Definition 5.1. The *Arithmetic Site* is the ringed topos $(\widehat{\mathbb{N}^\times}, \mathcal{O})$ consisting of the presheaf topos $\widehat{\mathbb{N}^\times}$ together with its structure sheaf $\mathcal{O} = \bar{\mathbb{N}}$.

Remark 5.2. The Arithmetic Site is a *ringed topos of characteristic 1*: its structure sheaf is a semiring with idempotent addition ($a \oplus a = a$), placing the entire construction in the world of tropical algebra rather than classical commutative ring theory. Connes and Consani describe it as the “algebraic geometric incarnation” of the non-commutative geometric approach to the Riemann hypothesis: the site provides a geometric framework, analogous to the function-field setting of the Weil proof, in which the relevant L -function — the Riemann zeta function — arises as a Hasse–Weil zeta function. The subsequent chapters make this precise by analysing the points of the site and the Frobenius correspondences acting on them.

Chapter 6

Points of the topos

In topos theory a *point* of a topos $\mathcal{E}$ generalises the notion of a point of a topological space: it is a geometric morphism $p : \mathbf{Set} \rightarrow \mathcal{E}$, pulling back each sheaf to its stalk. For a presheaf topos $\widehat{C}$, points correspond to flat functors $C \rightarrow \mathbf{Set}$. For $\widehat{\mathbb{N}^\times} = \widehat{B\mathbb{N}^\times}$ this means a point is a *flat $\mathbb{N}^\times$ -set*: a set X with a $\mathbb{N}^\times$ -action that arises as a filtered colimit of free $\mathbb{N}^\times$ -sets. The remarkable theorem of Connes–Consani is that such sets are classified, up to isomorphism, by the adèle class space of $\mathbb{Q}$ — a non-commutative space that already appears in Connes’ spectral approach to the Riemann hypothesis.

Definition 6.1. A *point* of a topos $\mathcal{E}$ is a geometric morphism $p : \mathbf{Set} \rightarrow \mathcal{E}$, i.e. an adjoint pair $p^* \dashv p_*$ where p^* is left exact.

Remark 6.2. For $\widehat{\mathbb{N}^\times}$, the flat $\mathbb{N}^\times$ -sets admit a concrete classification. A $\mathbb{N}^\times$ -set X is flat if and only if it is isomorphic (as an ordered $\mathbb{N}^\times$ -set) to a non-trivial subgroup of $(\mathbb{Q}, \mathbb{Q}_+)$ with the action of $n \in \mathbb{N}^\times$ by multiplication. The isomorphism classes of such subgroups are in natural bijection with elements of the double quotient $\mathbb{Q}_+^\times \backslash \mathbb{A}^f / \widehat{\mathbb{Z}}^*$: an adèle $a \in \mathbb{A}^f$ determines a subgroup $H_a = \{q \in \mathbb{Q} \mid qa \in \widehat{\mathbb{Z}}\}$ of $\mathbb{Q}$, and two adèles give isomorphic subgroups precisely when they lie in the same double coset. The rational adèles $a \in \mathbb{Q}^\times$ yield the single isomorphism class $\mathbb{Z}$, which is why the purely rational part of the space corresponds to the “points over $\mathbb{B}$ ” in the sense of Connes–Consani.

Theorem 6.3. *The space of isomorphism classes of points of $\widehat{\mathbb{N}^\times}$ is canonically isomorphic to the double quotient*

$$\mathbb{Q}_+^\times \backslash \mathbb{A}^f / \widehat{\mathbb{Z}}^*,$$

a piece of the adèle class space of $\mathbb{Q}$. More precisely, the category of points of $\widehat{\mathbb{N}^\times}$ is equivalent to the category of totally ordered groups isomorphic to non-trivial subgroups of $(\mathbb{Q}, \mathbb{Q}_+)$, with injective order-preserving homomorphisms as morphisms.

Proof. This is Theorem 2.2 of Connes–Consani (2014). Points of $\widehat{\mathbb{N}^\times}$ correspond to flat $\mathbb{N}^\times$ -sets, which are identified with non-trivial subgroups of $(\mathbb{Q}, \mathbb{Q}_+)$ by a direct analysis of the flatness condition. The double quotient description then follows from the identification $H_a = \{q \in \mathbb{Q} \mid qa \in \widehat{\mathbb{Z}}\}$ described in the remark above. (sorry) $\square$

Chapter 7

The Frobenius correspondences

The Frobenius correspondences are the analogue, in characteristic 1, of the Frobenius endomorphisms of a variety over a finite field. They do not act on the Arithmetic Site itself — that would require endomorphisms of the ringed topos $(\widehat{\mathbb{N}^\times}, \mathcal{O})$ — but rather on its *square* $(\widehat{B\mathbb{N}^{\times 2}}, \bar{\mathbb{N}} \otimes_{\mathbb{B}} \bar{\mathbb{N}})$, where $\mathbb{B} = (\{0, 1\}, \max, \times)$ is the Boolean semifield. The square is obtained by taking the product topos $\widehat{\mathbb{N}^\times} \times \widehat{\mathbb{N}^\times}$ and equipping it with the tensor product of structure sheaves.

For each pair $(n, m) \in \mathbb{N}^\times \times \mathbb{N}^\times$, Connes and Consani define an endomorphism $\text{Fr}_{n,m}$ of $\bar{\mathbb{N}} \otimes_{\mathbb{B}} \bar{\mathbb{N}}$ by

$$\text{Fr}_{n,m} \left(\sum_i q^{a_i} \otimes q^{b_i} \right) := \sum_i q^{na_i} \otimes q^{mb_i},$$

scaling the two factors independently. The Frobenius correspondence at a rational $\lambda = n/m \in \mathbb{Q}_+^\times$ (with n, m coprime) is then $\Phi_\lambda := \mu \circ \text{Fr}_{n,m}$, where $\mu : \bar{\mathbb{N}} \otimes_{\mathbb{B}} \bar{\mathbb{N}} \rightarrow \bar{\mathbb{N}}$ is the product map $\mu(q^a \otimes q^b) = q^{a+b}$. For general $\lambda \in \mathbb{Q}_+^\times$ the correspondence is defined by the formula

$$F(\lambda, q) \left(\sum_i q^{n_i} \otimes q^{m_i} \right) = q^{\inf_i (\lambda n_i + m_i)},$$

and extended to irrational λ by diophantine approximation.

Definition 7.1. For each $\lambda \in \mathbb{Q}_+^\times$, the *Frobenius correspondence* Φ_λ is the reduced correspondence on the square of the Arithmetic Site defined by the formula $F(\lambda, q)$ above. It acts on the points of $(\widehat{\mathbb{N}^\times}, \mathcal{O})$ over $[0, 1]_{\max}$ by the scaling map $x \mapsto \lambda x$ on the underlying adèle class space.

Theorem 7.2. *The Frobenius correspondences compose multiplicatively: $\Phi_\lambda \circ \Phi_{\lambda'} = \Phi_{\lambda\lambda'}$ for all $\lambda, \lambda' \in \mathbb{Q}_+^\times$ (with a canonical correction term when λ, λ' are both irrational and $\lambda\lambda' \in \mathbb{Q}_+^\times$). In particular, $\lambda \mapsto \Phi_\lambda$ is a homomorphism from $(\mathbb{Q}_+^\times, \times)$ into the composition monoid of correspondences on the square.*

Proof. This is Theorem 4.3 of Connes–Consani (2014). The composition law follows from the formula for $F(\lambda, q)$ and the identification of the range semirings $R(\lambda)$ described in their Section 3. (sorry) $\square$

Chapter 8

Connection to the Riemann zeta function

The Frobenius correspondences Φ_λ act on the points of the Arithmetic Site over $[0, 1]_{\max}$, which are parametrised by the adèle class space. Counting the fixed points of Φ_λ as λ varies produces a *counting distribution* $N(u)$, and the zeta function assembled from $N(u)$ via the Hasse–Weil formula turns out to be the complete Riemann zeta function. This is the central result of Connes–Consani and the main motivation for the entire construction.

The precise mechanism runs through Connes’ distributional trace formula. If $G = \mathrm{GL}_1(\mathbb{A}_{\mathbb{Q}})/\mathrm{GL}_1(\mathbb{Q})$ is the idèle class group and $\vartheta_u \xi(x) = \xi(u^{-1}x)$ is its scaling action on functions on the adèle class space, then for test functions $h(u) = g(|u|)$ supported away from 1 the trace formula gives

$$\mathrm{Tr}_{\mathrm{distr}} \left(\int_G h(u) \vartheta_u d^*u \right) = \sum_{v \in \Sigma_{\mathbb{Q}}} \int_{Q_v^*} \frac{h(u^{-1})}{|1 - u|} d^*u,$$

where the sum runs over all places of $\mathbb{Q}$. The counting distribution $N(u)$ is read off from the geometric side of this formula, and the zeta function is recovered by a standard Mellin-type transform.

Theorem 8.1. *The zeta function ζ_N defined by*

$$\frac{\partial_s \zeta_N(s)}{\zeta_N(s)} = - \int_1^\infty N(u) u^{-s} d^*u$$

from the counting distribution $N(u)$ of the Frobenius action on points of $(\widehat{\mathbb{N}^\times}, \mathcal{O})$ over $[0, 1]_{\max}$ is the complete Riemann zeta function $\zeta_{\mathbb{Q}}(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$.

Proof. This is Theorem 2.7 of Connes–Consani (2014). The counting distribution $N(u)$ is identified via Theorem 2.6 with the geometric term of Connes’ trace formula applied to the scaling action of $\mathbb{R}_+^*$ on the adèle class space, and the result then follows from the computation in Connes–Consani (2010). **(sorry)** $\square$